

18 Jun 2026 08:42:36 ET | 13 pages

Japan Chemicals

AI materials: Latest tech trends, Japanese company strengths and issues

CITI'S TAKE

We heard this week about AI-related materials technology trends in interviews with Mitsubishi Gas Chemical's Toru Harada (Executive Officer, Division Director, Electronics materials Division, Specialty Chemicals Business), Tsuyoshi Yamazaki (General Manager, Laminate Material Group, Electronics materials Division, Specialty Chemicals Business), and Kiyonari Hiramatsu (General Manager, Raw-New Materials Sales & Marketing Department, Electronics materials Division, Specialty Chemicals Business); Nitto Boseki's Hiroki Kajikawa (Senior Executive Officer, Deputy Division General Manager of the Corporate Management Division, General Manager Corporate Communication); JX Advanced Materials' Tomoji Mizuguchi (Senior Executive Officer, Deputy General Manager, Technology Group); Asahi Kasei's Toshiyasu Horie (Representative Director, Primary Executive Officer, CFO); and Mitsui Kinzoku's Kenji Kubou (General Manager, Business Planning Department) and Kenji Mori (Assistant General Manager, Marketing Group, New Product & Process Development). We also followed up on AGC's June 2 [semiconductor-related business briefing](#) and attended talks on PTFE at JPCA 2026 (June 10-12).

Strength—The emphasis on the strength of AI-related demand at this management level (not only on the IR side) was reassuring. Our meetings also confirmed the technological strength of Japan's materials industry. Such highly competitive Japanese firms can be considered the support base for current AI supply chains.

Issues: pricing strategy — However, our regular conversations with investors suggest stock market frustration with Japanese materials makers for a low-key approach to price negotiations. While our June 17 Macro to Micro report ([Japan Multi-Asset - The new pricing era: Pass-throughs fueling Japan's virtuous cycle](#)) notes signs that Japanese firms are becoming more proactive on price pass-through, we acknowledge that materials makers still look conservative in pricing strategy in comparison with foreign counterparts. Given tight supply/demand and involvement in key materials for AI supply chains (as well as the broad social acceptance of price pass-through signaled by the report) we think materials makers could show greater hunger for profit. In this vein we think JX Advanced Materials might take price hikes to a new dimension in upcoming negotiations, having announced [up to ten-fold expansion in InP substrate capacity](#) on June 16. We have heard particularly positive responses from foreign investors, who have referred to untypical aggressive capacity expansion and pricing strategy direction for a Japanese company.

Issues: ability to grasp downstream trends —Many business division managers indicated that upstream materials involvement means high sensitivity to primary customer demand trends but makes it difficult to discern demand and technological trends in the segments from secondary to final customers (sometimes the final application breakout is unknown, and the reasons underlying the magnitude of growth in demand are not clearly understood). Improvements to capabilities for gauging downstream trends in supply chains should facilitate early moves on next-generation product development and accurate assessment of underlying demand (avoidance of oversupply risk inherent in order duplication).

Yuta Nishiyama^{AC}

+81-3-6776-4643

yuta.nishiyama@citi.com

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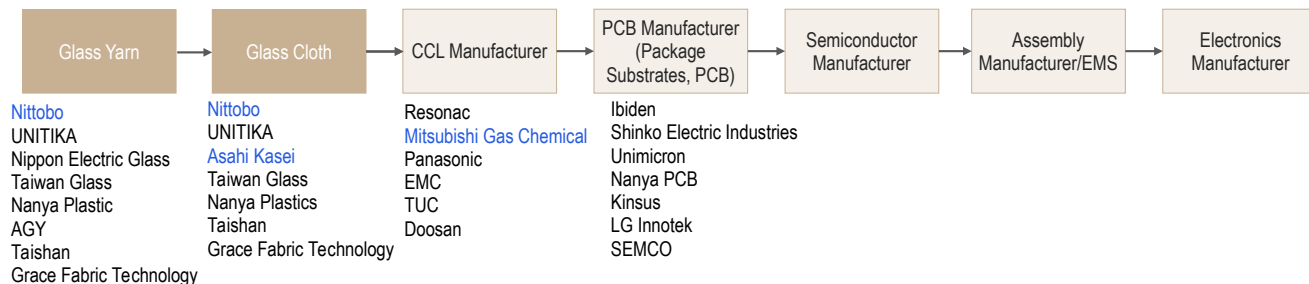
- **Mitsubishi Gas Chemical** — The favorable progress in customer evaluations of RS material, a new BT material product, is a positive. We got the impression that there is a very high likelihood the material will be adopted for use in FC-BGA packaging (especially for GPUs). MGC's materials were never seen as inferior to competing products in terms of key physical characteristics (like low-CTE, low-Dk), but we understand the semiconductor industry was generally reluctant to use alternative materials (due to lack of track record). However, the company explained that, in the last one-to-two years, efforts to find alternative materials have intensified across the supply chain amid CCL supply shortages (stemming originally from tighter low-CTE glass supply/demand), and that BT looks increasingly likely to be chosen as an alternative (given that its physical characteristics are basically superior). MGC intends to use the combination of RS and e-glass to target market share in applications like switches and PMICs, and the combination of RS and low-CTE glass to target share in BGA applications. In low-CTE glass, it appears the company is making progress in securing/using alternative suppliers beyond the main players. The company's OPE product currently offers low-Dk and low-Df characteristics in line with those of PPE in current target domains, but MGC is developing new products, aiming at characteristics on a par with hydrocarbon and PTFE materials. Shipments of samples have already started.
- **Nittobo** — The supply/demand balance is tightest for T-glass (low-CTE), followed by NER-glass (second-gen low-Dk), then NE-glass (first-gen low-Dk). Nittobo believes it is effectively the only company capable of supplying T-glass. Prospects of further increases in demand mean Nittobo may need to look at further capacity expansion; we understand the company would plan to enter into price negotiations as needed in that scenario. The NER-glass product cycle is proving more prolonged than initially expected, suggesting that the size of the addressable market could increase significantly. Specifically, it was originally thought that M9 and more advanced CCLs would require Q-glass or third-generation low-Dk glass (NEZ-glass), but we understand that recent advances in resin technology are generally seen as implying that the low-Dk requirements for glass cloth may not be as stringent as first thought (i.e., NER-glass may be sufficient). In low-Dk glass, overseas competitors have been making up ground sooner than expected, but management commented that their catching-up was not necessarily unfavorable from an overall market perspective given that the market was expected to expand beyond the size where Nittobo alone could handle all the supply. On the question of whether PTFE could threaten the low-Dk glass market, the company commented that the hurdle to adoption would be high given issues like PTFE's lack of rigidity.
- **JX Advanced Metals** — On June 16, the company [announced](#) plans to increase InP substrate production capacity by up to 10x through FY3/31. Management commented that, while InP substrate demand growth was initially projected at around 25% annually, recent momentum suggests growth of 30-40% could be expected. As of 2026, data centers are already using optical communication between racks, within data centers, and between centers. From 2028 onward, there could also be a transition from electrical to optical transmission within servers and within racks. This would mean higher demand for optical transceivers and, by extension, for InP substrates. Although JX announced a 10x production capacity increase, we do not think it would be surprising if demand increased by a greater factor. Citing a desire to quickly recoup capex (around ¥145bn cumulatively), the company also noted the importance of pricing strategy (our view is that, based on capex to date and the extent of price hikes JX has implemented in the past, there could be negotiations geared toward price increases of 2-3x). The company also explained that, with future GPU models

expected to have TDPs (thermal design powers) around 3x those of current models and their maximum power consumption expected to be 8x higher or more, the need to secure suitable power supply could lead to explosive growth in demand for tantalum powder for use in tantalum capacitors. The company also demonstrated how its magnetic target materials for HDD production have an edge in HAMR (heat-assisted magnetic recording) applications, with a share higher than the 50–60% share for PMR (perpendicular magnetic recording) applications.

- **AGC** — The company provides CCLs (marketed under the Meteorwave brand) for PCB manufacturing (customers are PCB manufacturers). Based on statements on AGC's website, we understand the resin used in the Meteorwave line is a PPE resin developed in-house. We understand that the cutting-edge ELL series within the Meteorwave line uses a hydrocarbon resin also developed in-house. The company says that use of these resins means that applications previously thought to require Q-glass now can use second-generation low-Dk glass cloth (this ties in with the explanation of the extended NER-glass product cycle in the Nittobo section). We understand evaluation of ELL series products is underway for applications like 1.6T (224Gbps x 8L) switches and routers. AGC is also independently developing low-Dk fluororesin, which it says is being well received by CCL manufacturer customers. We understand that, although this resin is not strictly PTFE, it has a molecular structure close to that of PTFE (per AGC's website). The material is said to share the low-Df, low-Dk characteristics of PTFE while also allowing lamination with other materials with no need for an adhesive layer (as we describe later, interlayer adhesion is a challenge with PTFE).
- **Asahi Kasei** — Asahi Kasei's PIMEL, used as a redistribution layer (RDL) material, has an extremely high share in cutting edge applications at key customers, with supply struggling to keep up with demand. The company plans to bring forward production increases. Demand for RDL materials is expected to increase as layer counts rise in line with the evolution of CoWoS packaging (CoWoS-S, CoWoS-R, CoWoS-L). In glass cloth production, the company expects to be able to secure yarn for L2 (second-gen low-Dk) cloth from new suppliers and to increase the volume of procurement. Sample shipments of low-CTE glass cloth have already started, with evaluations underway. The plan is to move to a full launch at end-FY3/27 or during FY3/28. Regarding Q-glass, while noting that customers would be motivated to use second-generation low-Dk glass to the extent possible (which squares with Nittobo's and AGC's explanations), management highlighted the edge offered by Asahi Kasei's combination of fiber and chemicals technologies.
- **Mitsui Kinzoku** — Beyond its established applications (like smartphone motherboards, memory), the MicroThin series has seen a sharp increase in inquiries for use in PCBs for optical transceivers. As for VSP, the sales weightings of HVLP4 and HVLP 5 are already quite high, with the firm commanding the top share in HVLP4 and an 80% share in HVLP5.
- **PTFE** — We attended a talk on PTFE at JPCA 2026 (Total Solution Exhibition for Electronic Equipment) held on June 10–12. PTFE's extremely favorable low-Dk and low-Df characteristics mean it is expected to be used in high-speed, high-volume, low-latency PCBs. However, there are issues with the use of PTFE, including a lack of rigidity, which leads to a tendency to warp (especially in large PCBs), low surface energy, which makes lamination difficult due to increased challenges with interlayer adhesion, and poor CTE characteristics, making heat deformation more likely (high coefficient of thermal expansion). To address

these challenges, we understand efforts are underway to improve rigidity and achieve low-CTE characteristics through measures like impregnating the substrate with fillers that have a low coefficient of thermal expansion.

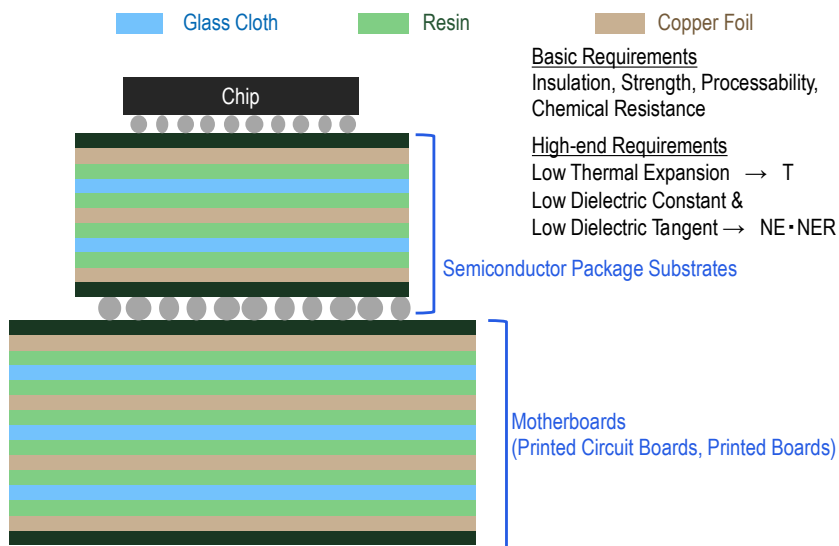
Figure 1. Glass cloth supply chain



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Source: Citi Research, Company data

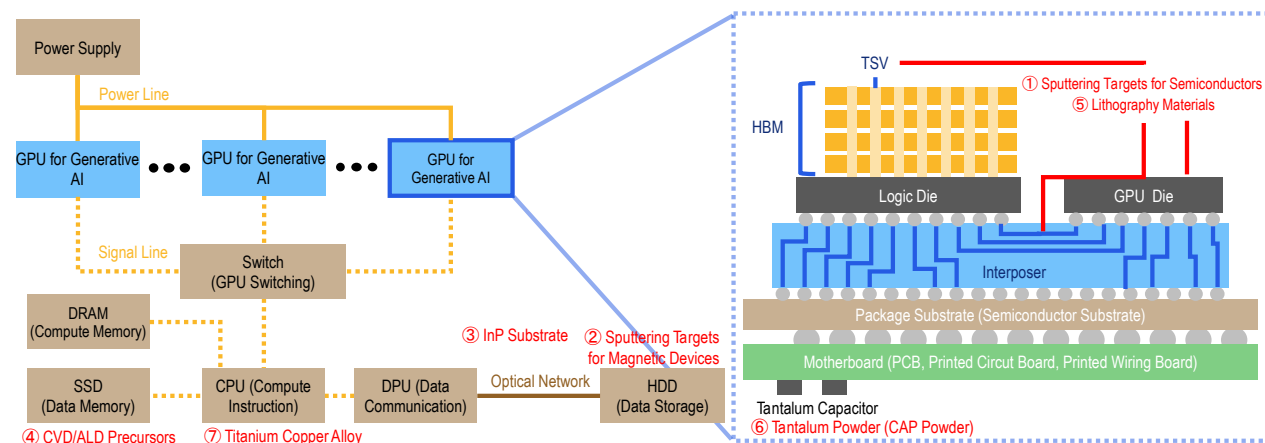
Figure 2. The structure of CCL



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Figure 3. JXAM: Product deployment in data centers (particularly AI servers)



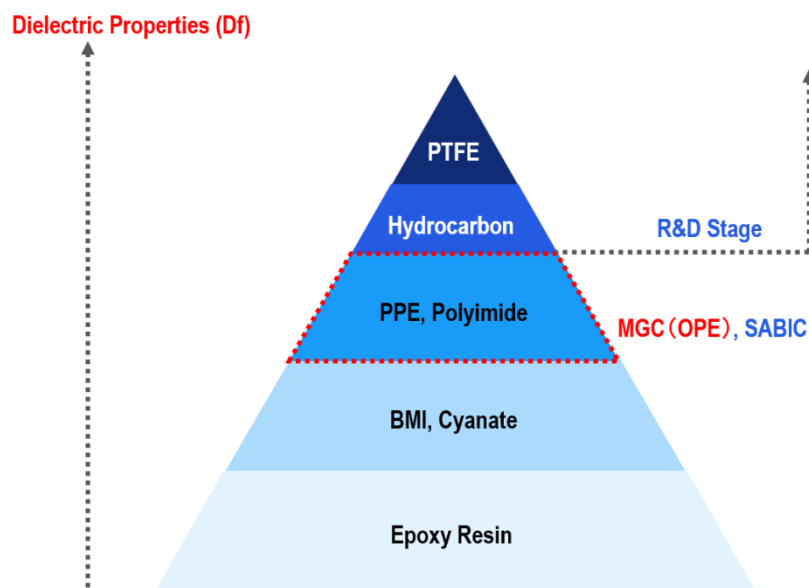
Product	Overview	Market Share	Competitor
① Sputtering Targets for Semiconductors	Metal materials for semiconductor circuit formation. Approximately 30% of sales are for leading-edge logic, 5% for interposer redistribution layers (RDL), and 5% for TSV formation in HBM.	approx. 60%	CN Kfmi US Honeywell
② Sputtering Targets for Magnetic Devices	Metal materials used to form HDD media, with 80–90% of sales for data centers.	approx. 60%	JP Tanaka Precious Metals
③ InP Substrates	Adoption of InP substrates for optical networks between AI servers (optoelectronic devices) is rapidly expanding due to high transmission speed, low loss, and superior high-frequency performance. About 80% of sales are for data-center applications.	approx. 40%	JP Sumitomo Electric US AXT
④ CVD/ALD Precursors	In highly stacked NAND (300 layers and above), molybdenum (Mo) is attracting attention as a new material for word lines. Sales remain limited as of 2025, but are expected to expand from 2026 onward.	approx. 30%	US Entegris FR Air Liquide
⑤ Lithography Materials	The company supplies reflective materials for EUV light in EUV exposure systems.	approx. 100%	NA
⑥ Tantalum Powder (CAP Powder)	Because AI servers tend to generate high heat, adoption of heat-resistant tantalum capacitors as power-delivery components is rapidly increasing, and approx. 40% of sales for CAP powder are for data centers.	approx. 50%	US GAM
⑦ Titanium Copper Alloy	While copper alloys had been used for connectors in general-purpose servers, they are rapidly being replaced by Titanium Copper due to heat-resistance and related requirements. Approx. 50% of Titanium Copper sales are for data center.	approx. 90%	JP DOWA Metaltech

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Note: All numbers are Citi Research estimates.

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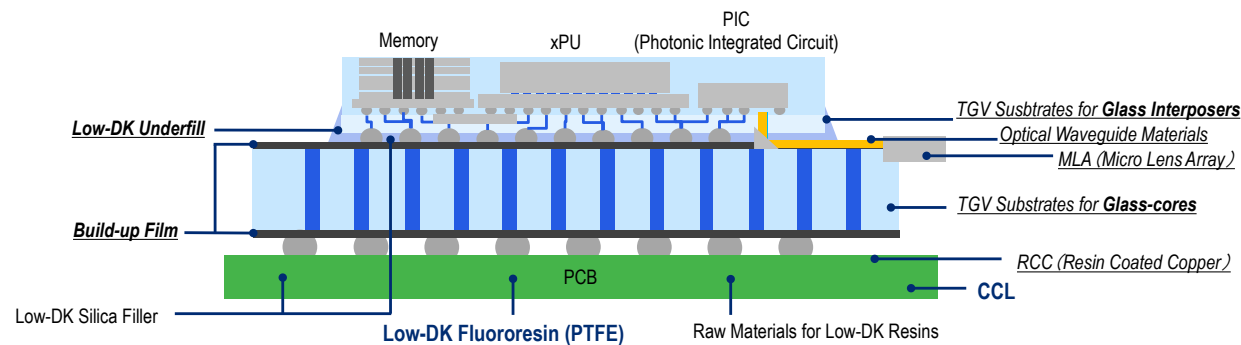
Figure 4. Resins with low dielectric properties and OPE application areas



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Source: Citi Research, Company data

Figure 5. AGC's solutions for semiconductor target applications



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Note: Underlined items are products under development.

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